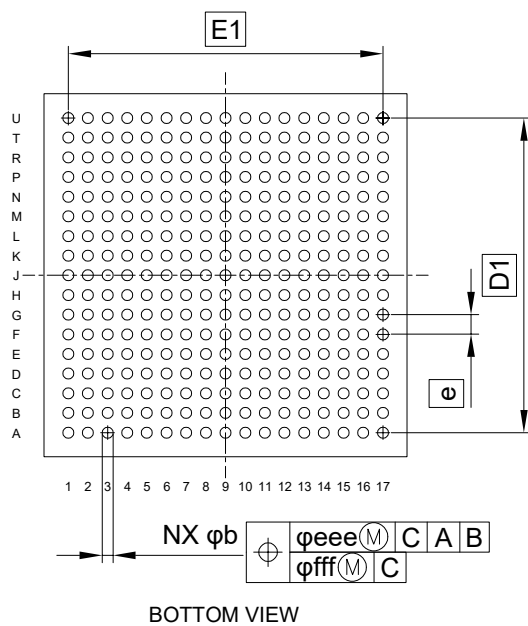
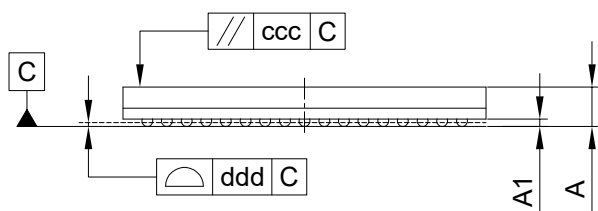
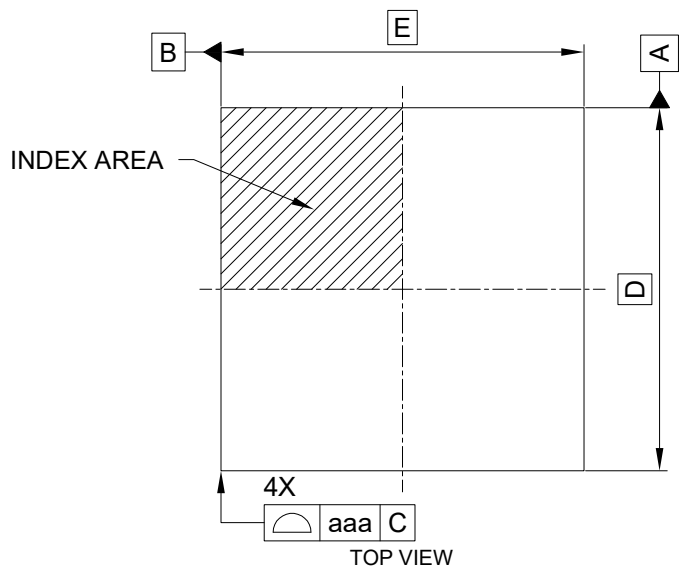


Appendix 2. Package Dimensions

Information on the latest version of the package dimensions or mountings is displayed in “Packages” on the Renesas Electronics Corporation website.

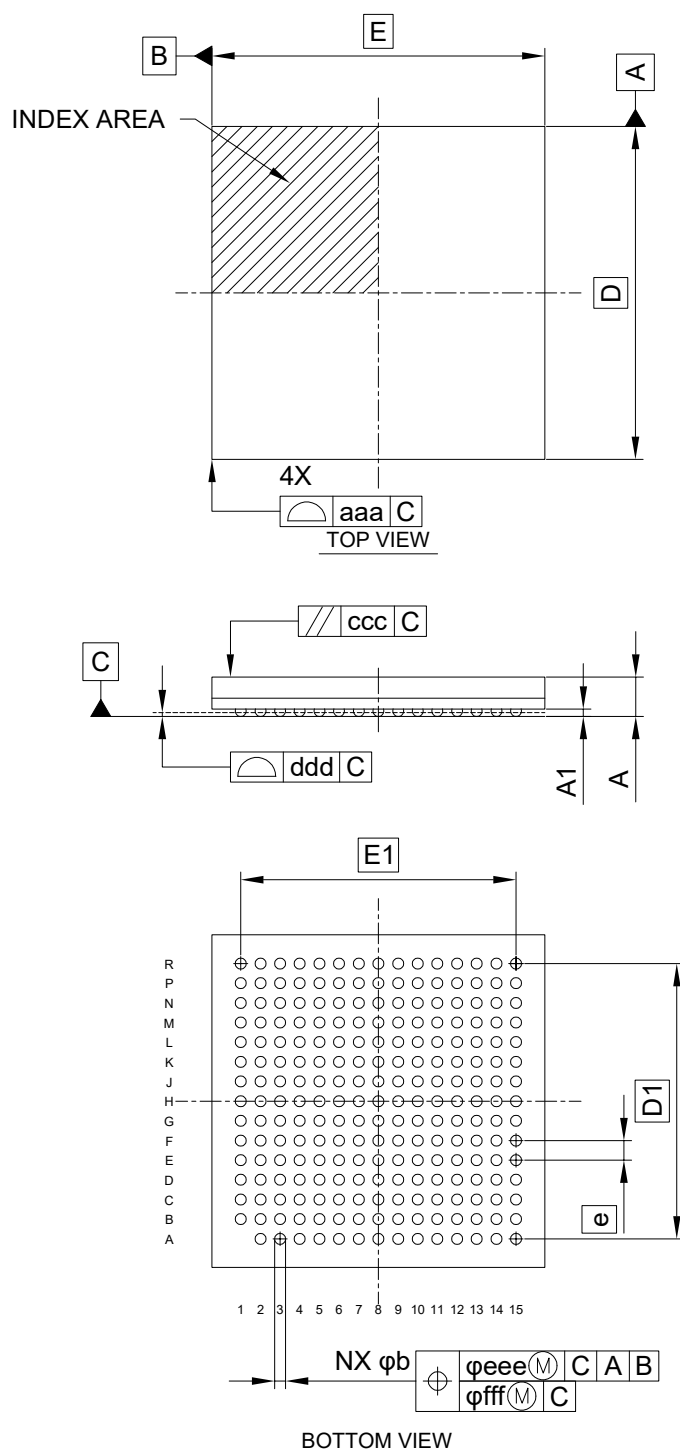
JEITA Package code	RENESAS code	MASS(TYP.)[g]
P-LFBGA289-12x12-0.65	PLBG0289JA-A	0.38



Reference Symbol	Dimension in Millimeters		
	Min.	Nom.	Max.
D	—	12.00	—
E	—	12.00	—
D1	—	10.40	—
E1	—	10.40	—
A	—	—	1.38
A1	0.20	—	—
b	0.31	0.36	0.41
e	—	0.65	—
aaa	—	—	0.15
ccc	—	—	0.20
ddd	—	—	0.10
eee	—	—	0.15
fff	—	—	0.08
N	—	289	—

Figure A2.1 BGA 289-pin

JEITA Package code	RENESAS code	MASS(TYP.)[g]
P-LFBGA224-11x11-0.65	PLBG0224JA-A	0.32



Reference Symbol	Dimension in Millimeters		
	Min.	Nom.	Max.
D	—	11.00	—
E	—	11.00	—
D1	—	9.10	—
E1	—	9.10	—
A	—	—	1.38
A1	0.20	—	—
b	0.31	0.36	0.41
e	—	0.65	—
aaa	—	—	0.15
ccc	—	—	0.20
ddd	—	—	0.10
eee	—	—	0.15
fff	—	—	0.08
N	—	224	—

Figure A2.2 BGA 224-pin

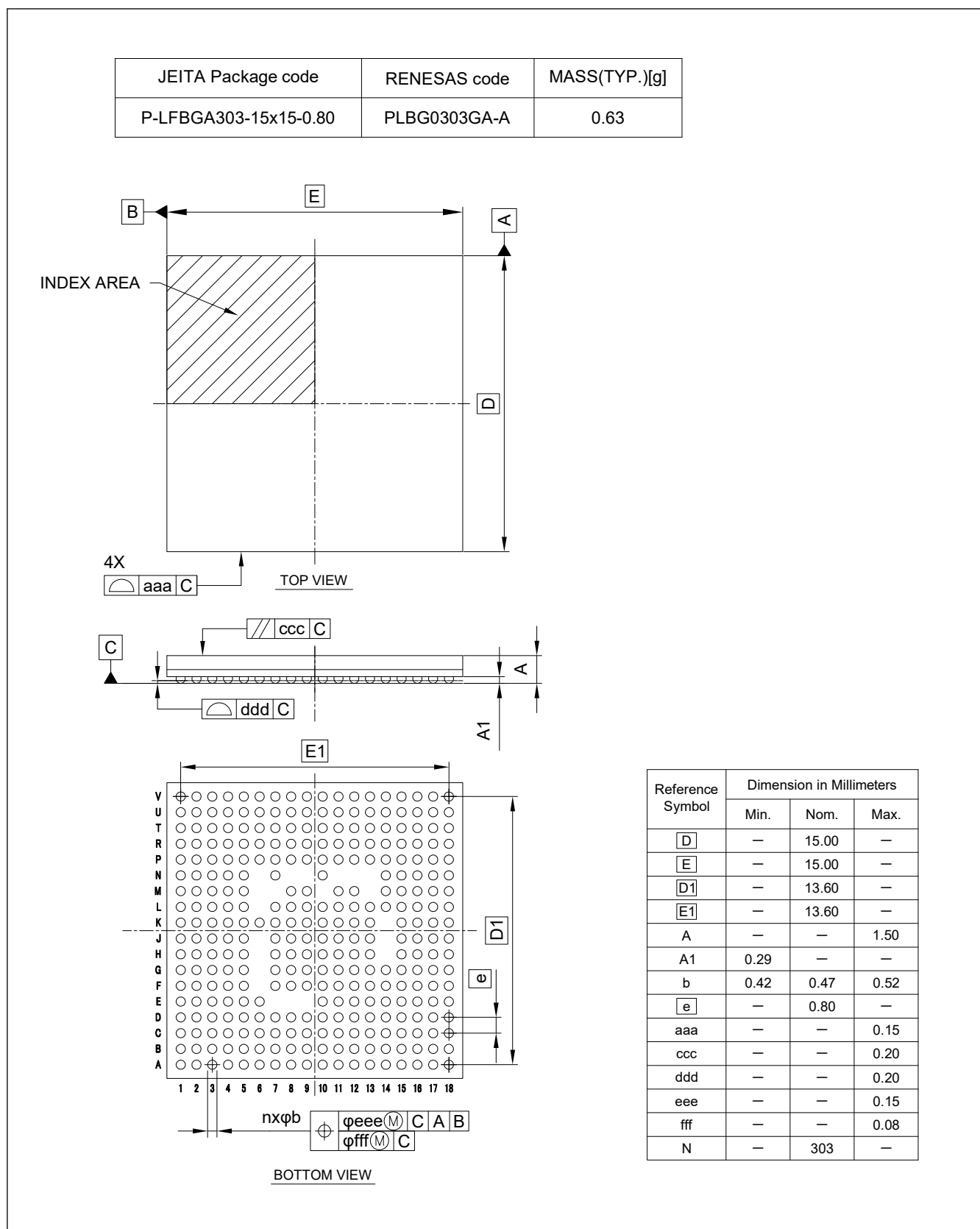


Figure A2.3 BGA 303-pin

JEITA Package code	RENESAS code	MASS(TYP.)[g]
P-HLQFP176-24x24-0.50	PLQP0176KK-A	1.90

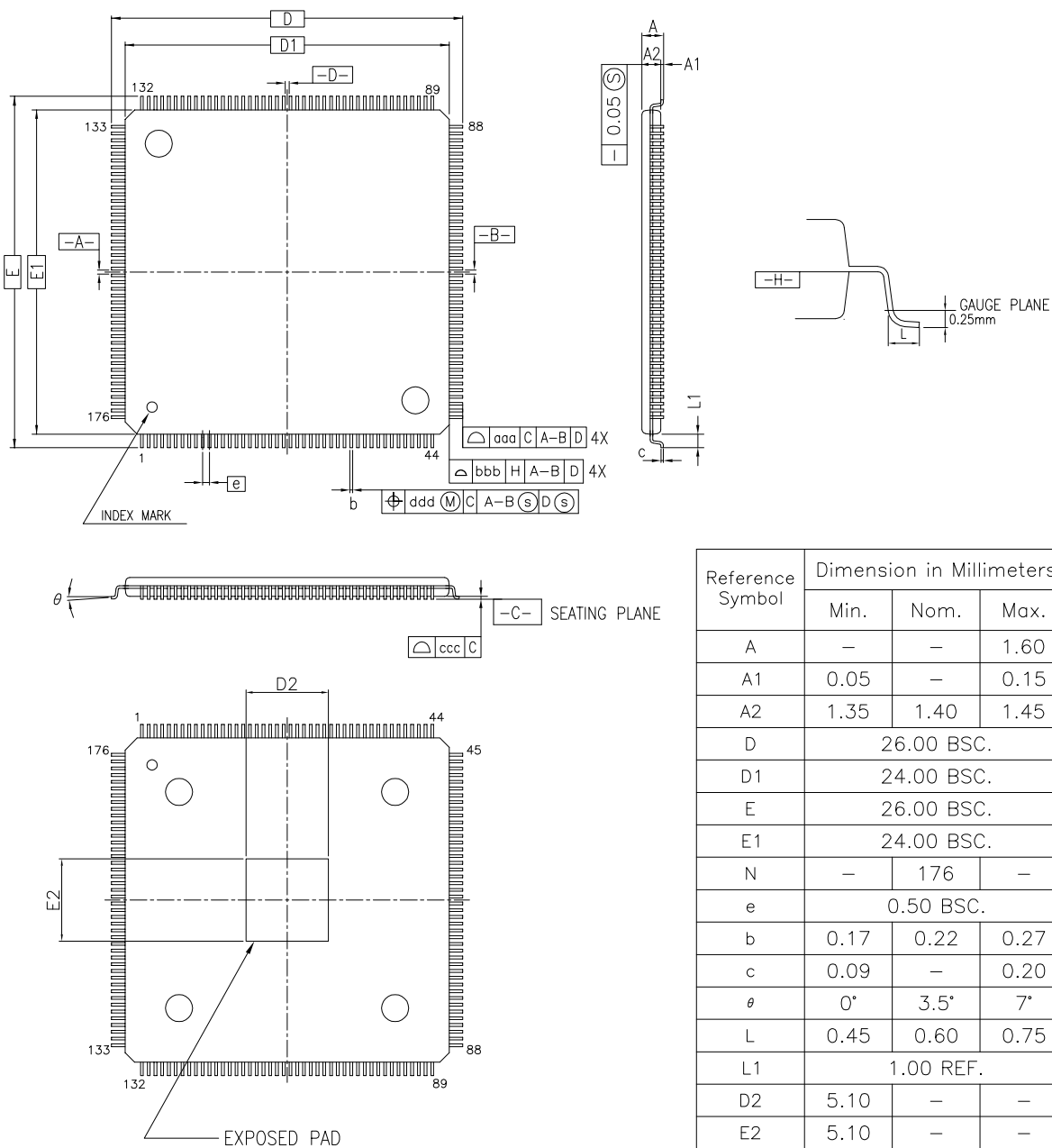


Figure A2.4 HLQFP 176-pin